SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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1. TEAM AND PROJECT OVERVIEW

Team Name

SDC4

Project / Solution Name

NEXTEP

Name (Leader)

YUE XIONGHE

Email & Phone (Leader)

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University Affiliation(s)

XIAMEN UNIVERSITY MALAYSIA

Live Prototype URL

https://nextep-ericyywies.replit.app

Code Repository URL (public / access granted to judges)

https://github.com/ERYCYUE/SDC_HACKATHON_SUBMISSION_TEMPLATE_GROUP_SDC4

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Our solution is built on a responsive cloud-native web architecture powered by AI reasoning and real-world global employment data:

1. User Interface : A modern interactive web frontend built and hosted via Codex, delivering an intuitive user experience for career path exploration and visualization.

2. AI Reasoning & Core Logic: The application leverages the DeepSeek API to analyze user profiles, process dynamic queries, and generate real-time personalized career path insights.

3. Data Layer: Integrates an open-source global recruitment and headhunting dataset, providing comprehensive data on global job market trends, skill requirements, and industry benchmarks for high-precision career matching.

4. Hosting & Deployment: Live-hosted at our site, ensuring fast, reliable, and accessible cloud delivery.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Next.js (React)	Full-stack web framework: frontend pages and API Routes backend	MIT
Node.js	Server-side runtime for the backend and API calls	MIT
Tailwind CSS	Styling of the user interface	MIT

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
DeepSeek API (DeepSeek)	Integrates a large language model to generate reasonable, personalised career-path plans for users	Paid
OpenDOSM API, data.gov.my (Department of Statistics Malaysia)	Malaysia labour force (monthly/quarterly), skills-related underemployment and wages by occupation, imported into the app database	Free (no key required)
data.gov.sg API (Ministry of Manpower, Singapore)	Singapore job vacancies, vacancies by occupation, occupational wages and employment, imported into the app database	Free (no key required)
BLS Public Data API (U.S. Bureau of Labor Statistics)	US labour force, occupational projections, wages and JOLTS job openings, imported into the app database	Free (optional free registration key)
O*NET Web Services (U.S. DOL); CareerOneStop Web API	US occupation tasks, skills, knowledge and training programmes, mapped to the app skill and role catalog	Free (free registration key / token)
Bundesagentur fur Arbeit Statistik API (Germany); ESCO API (European Commission)	Germany registered job vacancies by region; ESCO occupation and skill taxonomy, mapped to the app catalog	Free

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
OpenAI Codex (OpenAI)	Used for development only (code generation and editing); not used for hosting	Free
Replit (SDC sponsor)	Cloud development, full-stack hosting and deployment of the prototype (Next.js frontend + API Routes backend)	SDC sponsor-provided (Replit cloud)
GitHub (GitHub, Inc.)	Public code repository for version control and judge access	Free
PostgreSQL (Replit)	Cloud database that stores and queries the recruitment and labour-market datasets for skill matching	SDC sponsor-provided (Replit cloud)

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
DeepSeek-V4.1-Flash (API: deepseek-flash)	DeepSeek, via DeepSeek API	Pre-trained; used directly through the API, no fine-tuning	DeepSeek Open Platform Terms of Service

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
BOSS Zhipin EEE job postings (2,639 unique jobs, 25 EEE directions in Shanghai, Shenzhen, Beijing, Chengdu; Oct 2026)	zhipin.com (BOSS Zhipin public job listings)	Self-collected from publicly available job postings (education, school, experience and salary requirements)	BOSS Zhipin platform content; used for non-commercial academic research only	N
Malaysia labour data: OpenDOSM (LFS, wages by occupation) and TalentCorp MyCOL	open.dosm.gov.my; talentcorp.com.my	Public (official statistics and reports)	Open data, attribution (CC BY 4.0)	N
Singapore labour data: MOM job vacancies and occupational wages; SkillsFuture Skills Framework	data.gov.sg; skillsfuture.gov.sg	Public (official statistics)	Singapore Open Data Licence	N
US labour data: BLS (employment, projections, wages, JOLTS) and O*NET skills and tasks	bls.gov; onetcenter.org	Public (official statistics)	Public domain (BLS); CC BY 4.0 (O*NET)	N
Germany and EU data: BA Statistik (vacancies, wages), ESCO skills taxonomy, Cedefop skills forecast	arbeitsagentur.de; esco.ec.europa.eu; cedefop.europa.eu	Public (official statistics and taxonomies)	Open, free use with attribution	N
UN Sustainable Development Goals and targets	sdgs.un.org/goals	Public	UN terms of use, with attribution	N
User skill profile	Entered by the user in the app	Self-provided by user at runtime	N/A	Y (stored locally in the user's browser only)

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
OpenAI Codex (model: GPT-6.1 Sol)	Building the website: generating and editing code, UI and page content	Major
Replit Agent / Replit AI	Assisting with full-stack code implementation, bug fixing, and cloud deployment configuration.	Moderate
ChatGPT / Gemini / Claude	Assisting with documentation polishing, prompt optimization, and pitch slide text structure.	Minor

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every piece of AI-generated code was reviewed by our team. Where we found weaknesses, we pointed them out and had the tool revise them until the result met our standards. We then tested the prototype hands-on, clicking through every page and feature to confirm it worked as intended. The data and written content were cross-checked by several different AI models and then verified manually by team members. We also examined DeepSeek's career-planning output to make sure its advice was reasonable and well grounded. Our team takes full responsibility for the final code, content and claims.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit (SDC sponsor)	Cloud platform: online IDE, full-stack hosting and deployment	Cloud development environment for Next.js + Node.js with live reload; hosts the frontend and API Routes backend; securely calls the DeepSeek API; deployed the judge-accessible prototype (nextep-ericyywies.replit.app)

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was essential to our prototype. It gave our team a zero-setup cloud environment for Next.js and Node.js, live reload for fast iteration, and one-click full-stack hosting with a built-in PostgreSQL database, which together turned our proposal into a working, judge-accessible prototype within a short hackathon timeline. It also let us call the DeepSeek API securely from the backend. In principle the code is standard and could be moved to another cloud provider, but this would require reconfiguring hosting, the database and deployment, and would cost valuable development time that Replit allowed us to invest in features instead.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Collection & Storage: User profiles, chat history, and preferences are stored locally in the browser via `localStorage` (`eee-next-state-v2`), while imported dataset files reside in IndexedDB (`eee-next-local`). No personal data or chat tables are stored on servers; server-side storage (Cloudflare D1) only holds public job market samples. Data persists locally until the user resets exploration, clears browser cache, or device cleanup occurs.

Third-Party AI Integration: DeepSeek API is disabled by default. Only when explicitly toggled per session by the user, minimal contextual data (current query, last 6 messages, structured profile, and career summaries) is forwarded. Raw uploaded CV files are parsed locally and never sent to the API.

Protection: The server acts strictly as a stateless proxy without logging request payloads, ensuring zero server-side personal data retention.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

LLM Hallucinations & Bias: Large language models may generate inaccurate career recommendations or display historical hiring bias. We mitigate this by using strict system prompt grounding, JSON-schema constraints, and validating skill extractions against structured, live recruitment datasets.

API Rate Limits & Latency: LLM response times can affect user experience. We implement response caching for common skill queries and asynchronous request handling to maintain smooth UI performance.

Provider Dependency: Relying solely on DeepSeek API poses uptime risks. Our backend architecture uses an abstraction layer, enabling quick fallback switches to alternative LLM APIs (e.g., OpenAI, Claude) if service disruptions occur.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☑ Any use of official SDC sponsor resources has been declared in Section 5.

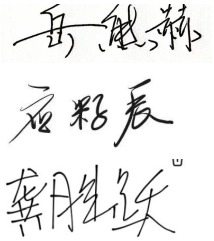
☑ The prototype described here is accessible to judges through the links provided in Section 1.

☑ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] 陈昊宇



[Date]

24/10/2026